Roll No.....

DRONACHARYA COLLEGE OF ENGINEERING, GURUGRAM
PRE-UNIVERSITY EXAMINATION APRIL, 2025
(Gurugram University)

Semester: IV (B Tech) Branch: CSE(AI&ML)/CSE(IOT)

OPERATING SYSTEM

Subject with Code: OPERATING SYSTEM

Time: 3 hours **Max. Marks: 70**

Note: *Q.1 is compulsory. Attempt any ONE question from each Unit. Each question carries of 14 marks. Assume data, if any missing.*

Q. 1. Objective type / Short-answer type questions. ***(2x7=14)***

i. Consider the following processes with their burst times and a time quantum of 4 ms Assume all processes arrive at time 0. **[CO1]**

Process-Id	Burst Time
P1	4
P2	7
P3	8

Calculate: Waiting Time (WT),Turnaround Time (TAT),Average Waiting Time(AWT)

ii. Discuss the services of OS? **[CO1]**

iii. What are the benefits of using threads? **[CO2]**

iv. Give the examples of batch processing? **[CO1]**

v. Define process management in OS? **[CO4]**

vi. What are the key features of file management? **[CO2]**

vii. Given a system with a 32-bit virtual address space and a page size of 4 KB, calculate the number of pages and the size of the page table if there are 4 GB of physical memory. **[CO3]**

UNIT-I

Q1 Given the following burst times for 4 processes, calculate the waiting time and turnaround time using SJF (Non-Preemptive) scheduling. Assume all processes arrive at time 0. **[CO1]**

Process-Id	Burst Time
P1	7
P2	2
P3	6
P4	3

Waiting Time (WT) and Turnaround Time (TAT) as defined above.

Q2 Following are the processes with burst times, calculate the Average Waiting Time and Average Turnaround Time using the Round Robin (RR) scheduling algorithm with a time quantum of 3 units. Assume all processes arrive at time 0. [CO2]

Process-Id	Burst Time
P1	1
P2	9
P3	4
P4	7

Time Quantum (TQ): 2 units

Find Waiting Time (WT) and Turnaround Time (TAT).

UNIT-II

Q3 a) Describe the Dining Philosophers Problem in terms of process synchronization. [CO2]

b) What are the two models of inter-process communication ? [CO2]

Q4. Explain the Banker's Algorithm for deadlock avoidance. [CO4]

UNIT-III

Q5 a) Explain the difference between physical memory and virtual memory. [CO3]

b) How does paging help in managing fragmented memory in a system? [CO3]

Q6 a) Given a series of page references (e.g., 6, 0, 1, 2, 0, 9, 0, 4, 4, 6), determine the page fault rate using the LRU page replacement algorithms (for a system with 4 page frames). [CO3]

b) What is the Optimal page replacement algorithm, and how is it different from FIFO and LRU? Why is it impractical in real systems? [CO3]

UNIT-IV

Q7 a) Explain the differences between process management in Unix and Windows. [CO3]

Q8 C-SCAN scheduling:

Start at position 50 and move right (toward 100).After reaching the highest request, jump to the other end (0) and continue servicing requests. Requests serviced in order:

$20 \rightarrow 50 \rightarrow 120 \rightarrow 200 \rightarrow 3 \rightarrow 20$.Total distance traveled? [CO3]